Gagan Thakur

Mandi, Himachal Pradesh

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Education

National Institute of Technology, Hamirpur — **Aug. 2024 – Present**
B.Tech in Electronics and Communication Engineering

Swami Vivekanand Sr. Sec. School — **Apr. 2010 – Apr. 2023**
High School — *Mandi, India*

Technical Skills

Languages:	Python, C++, Rust, JavaScript, C
ML / DL Frameworks:	TensorFlow, PyTorch, Keras, scikit-learn, XGBoost
Research Areas:	Deep Learning, NLP, Computer Vision, Multimodal AI, Reinforcement Learning
Data & Processing:	NumPy, pandas, OpenCV, Matplotlib, Seaborn
MLOps & Deployment:	Git, DVC, MLflow, FastAPI, Docker, Streamlit, CI/CD, AWS

Work Experience

Author & Developer: *Alphabets of Machine Learning* — **Apr. 2025 – Present**
Self-Driven Open-Source Project — *Remote*

- Designed and published **26 end-to-end ML projects** spanning supervised, unsupervised, and reinforcement learning with full documentation and reproducible code — used as a learning resource by peers.
- GitHub: github.com/seika-afk/Alphabets-of-Machine-Learning

Projects

NeuroBlock – Neural Network Library from Scratch [GitHub] — *C++*

- Implemented a fully connected ANN library in C++ from scratch without any external ML framework, covering custom tensor operations (matmul, transpose, reshape, He/Xavier initialization), forward propagation, backpropagation with correct gradient computation, and epoch-based training with loss tracking.
- Implemented activation functions (ReLU, Sigmoid) and loss functions (MAE, BCE) from first principles, demonstrating deep understanding of neural network internals.

Visual Question Answering – Multimodal AI [GitHub] — *Python, TensorFlow, Keras, OpenCV, scikit-learn*

- Built a multimodal VQA pipeline fusing CNN-based image features with tokenized question embeddings to predict answers over a multi-choice answer space using categorical cross-entropy.
- Designed a custom `DatasetPipe` for joint image–text preprocessing, tokenization, and TensorFlow dataset batching; used callbacks (EarlyStopping, ReduceLROnPlateau, ModelCheckpoint) for stable training.

nl2bash – Natural Language to Bash via Transformer [GitHub] — *Python, PyTorch, Transformer, seq2seq*

- Trained a seq2seq Transformer from scratch to translate natural language instructions into executable Bash commands, implementing multi-head attention, positional encoding, and encoder-decoder architecture without high-level abstractions.
- Conducted extended training runs (1000+ epochs) to study convergence behaviour on the NL2Bash dataset, exploring the effect of training duration on sequence generation quality.

Achievements

- Solved **LeetCode Top 150** series twice, reinforcing strong foundations in data structures and algorithms under constraints.

Interests

- Deep learning theory and neural architecture research
- Digital art and creative visual design
- Literature and story writing